

IVP PROJECT REPORT

NTIRE 2026 Image Shadow Removal Challenge (@ CVPR 2026)

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Project Description:

This project focuses on the task of image shadow removal, which involves eliminating cast shadows from images while preserving the original scene structure and visual consistency. Shadows often introduce strong and spatially varying illumination changes that can negatively affect both visual quality and the performance of computer vision systems.

The objective of shadow removal is to generate shadow-free images that appear perceptually realistic and remain faithful to the original content. This problem is particularly challenging due to the wide variability in shadow characteristics, including differences in shape, softness, intensity, and their interaction with diverse textures and materials.

Developing robust shadow removal techniques is important for improving the reliability of higher-level computer vision tasks such as object detection, semantic segmentation, tracking, and scene understanding. It also has practical significance in applications like image editing, autonomous driving, and robotic perception, where consistent visual representation is critical.

Although recent deep learning approaches have achieved notable progress, ensuring strong generalization across varying scenes and lighting conditions remains an ongoing challenge addressed in this project.

Dataset:

Training:

- **1000 pairs of images**
- **Shadow Image & Shadow-Free Image (Ground Truth)**
- **Image Resolution: 960 × 720 pixels**
- **Format: RGB images**



Validation:

- 75 images
- Shadow-only images (no ground-truth provided)
- Resolution: 960 × 720
- Contains real-world shadow variations



Test:

- 75 images
- Shadow-only images (no ground-truth provided)
- Resolution: 960 × 720
- Contains real-world shadow variations



Our Approach:

In this project, we adopt a lightweight deep learning-based approach for image shadow removal using a simplified encoder–decoder style convolutional network inspired by transformer-based restoration models.

Model Architecture

We design a compact network named **LiteRestormer**, which focuses on efficiency while maintaining reasonable restoration performance. The model consists of three main components:

- An **input convolution layer** that projects the RGB image into a higher-dimensional feature space.

- A sequence of lightweight residual blocks (**LiteBlocks**), each containing two convolutional layers with GELU activation and a residual connection to preserve image details and stabilize training.
- An **output convolution layer** that maps the processed features back to a 3-channel RGB image.

Unlike full-scale transformer-based architectures, our model avoids heavy attention mechanisms and instead relies on stacked convolutional residual learning, making it computationally efficient and suitable for faster inference.

Dataset Handling

We use the NTIRE shadow removal dataset, where each sample consists of:

- An input image containing shadows (`_in`)
- A corresponding ground truth shadow-free image (`_gt`)

The images are loaded using OpenCV, normalized to the range $[0, 1]$, and converted into PyTorch tensors. A custom dataset class is implemented to efficiently handle paired data loading.

Training Strategy

The model is trained using the following setup:

- **Loss Function:** L1 Loss (Mean Absolute Error), chosen for its effectiveness in preserving image sharpness and reducing blurring.
- **Optimizer:** AdamW optimizer with a learning rate of 2×10^{-4} for stable convergence.
- **Batch Size:** 1 (to handle high-resolution images within memory constraints).
- **Epochs:** 20 training epochs.
- **Mixed Precision Training:** Automatic Mixed Precision (AMP) with gradient scaling is used to improve training speed and reduce GPU memory usage.

Inference and Output Generation

During inference, the trained model is applied to validation images:

- Input images are normalized and passed through the network.
- The output is clamped to valid pixel ranges and converted back to image format.
- Final shadow-free images are saved for evaluation and submission.

Efficiency Considerations

The proposed approach emphasizes computational efficiency:

- A lightweight architecture reduces model complexity.
- Mixed precision accelerates training and inference.
- The method achieves a runtime of approximately **0.43 seconds per image** on GPU.

Leaderboard:

Validation:

32	kannappan	2026-02-25 21:49	564580	20.263	0.179	0.7568	-1.0	-1.0	-1.0
34	kesavram	2026-02-24 18:26	561248	19.9	0.172	0.761	0.43	0.0	0.0

Test:

46	kesavram	2026-03-16 18:52	629840	19.889	0.1739	0.7623	0.43	1.0	1.0
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Conclusion:

In this project, we developed a lightweight model for image shadow removal using a simplified **LiteRestormer** architecture. The approach focused on efficiency and stable training using an L1 loss function.

Our model achieved a PSNR of **19.889 dB**, compared to the top score of **26.912 dB**, indicating a noticeable performance gap. This is mainly due to the use of a simplified architecture without advanced components like attention mechanisms or multi-scale feature extraction.

Despite this, the model produced reasonable results with low computational cost and fast inference. Future improvements could include incorporating more advanced architectures, improved loss functions, and better training strategies to enhance performance and generalization.